

- 1) The difference between classification and regression is that they differ in their output. Classification produces discrete labels, and regression produces a continuous set of values. Classification can categorize data into different groups, while regression models a relationship to predict a specific quantity. Classification uses accuracy, precision, and F1, and regression utilizes mean squared and r squared.
- 2) The K mean is an unsupervised cluster algorithm that groups data points into clusters and mean line to distinguish between distinct classes.

First a K value is chosen, assign data points to the K value line using the Euclidean distance.

Calculate the mean of the data points and adjust the centroid with the K value to get the new mean.

Repeat the K value assignment and update until the centroids do not change anymore.

The larger the K value the more generalized the smaller the k value the more specific it can be.

- 3) Predicting house prices off square footage is better suited for linear regression because the price of the house and sqft are continuous. And so if the sqft price increase typically the house price would also.

Predicting if a customer will purchase something is better suited for logistic regression. I say this because it is a binary function and there is no reason to go beyond 1 when the output should only be 1 or 0.